How TiRex makes a forecast

NX-AI · 35M parameters · xLSTM (sLSTM) recurrent network · zero-shot

1. One series at a time

TiRex reads only the target's own past. A ? is a value not known yet.

past *future*

Target temperature

■■■■■■■ ? ? ?

Covariates such as day length are not used.

Five metal prices become five separate forecasts.

2. Scale, then cut into patches

The context is scaled by its own mean μ and standard deviation σ. Gaps stay as missing values.

$$\tilde{x} = \frac{x - \mu}{\sigma}$$

value seen?

one patch = 32 steps

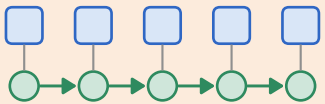
residual MLP

one token: 512 numbers

3. Twelve recurrent blocks

No attention: an sLSTM cell reads the patches in order and carries a state from one to the next. RMSNorm in front, shortcut around.

patch tokens



state carried forward

sLSTM: 4 heads, exponential input and forget gates, a state that tracks the series

Feed forward: gated (SwiGLU), width 1,408

4. 32 steps per pass

After a final RMSNorm, an output network turns the last token into **9 quantiles** for each of the next **32 steps**.

median

60% band

80% band

For a longer horizon, the forecast patch goes back in as missing values, never as numbers, and the next one is predicted. The median is the point forecast.

A recurrent state instead of attention: the cost grows only linearly with the context.

12 blocks · width 512 · 4 heads · patches of 32 in and out · trained on 2,048-step contexts

In the notebooks: last 2,048 days or full monthly history · each metal on its own · no covariates · a 90-day horizon takes 3 passes